DRS3

Measurements Round-Trip Delay

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1 Method

1.1 Description of the Measurement Method

The following method and steps were performed to gather measurement data.

1. Master node software transmits a small packet in a loop. The next package is sent when the previous came back. This was used to fulfill R-RT02.
2. Slave node receives the data and immediately transmits it back (R-RT03).
3. The master records the time from transmission to reception of the same packet (round-trip time) in a memory array (R-RT04). The array was exported to an CSV file.
4. A fixed timeout was used to in case no data was received (R-RT06).

1.2 Description

A Raspberry Pi acts as the master, while a ESP32 is a UDP echo slave. A 32-byte packet is echoed back unchanged each time. Only one request is active at any given time. After receiving a valid response, the next request starts without any intentional delay. Following a timeout, the process continues with a new sequence number; there are no retries for the same request. UDP was chosen to ensure that packet loss remains explicitly visible. In case of an Error the nodes will not retry the request, but instead continue with the next sequence number. This is to ensure that the round-trip time is not influenced by retries. This error handling was implemented to fulifll requirement R-RT06. The master node records the round-trip time for each packet and stores it in an array. After a predefined number of measurements, the data is exported to a CSV file for further analysis.

1.3 Measurement Setup



Abbildung 1: Schematic measurement setup.



Abbildung 2: Picture of the real setup showing every component connected.

2 Results

In this section the results of the measurements with the specified conditions are presented. The measurements were performed with a fixed timeout of 1000 ms.

2.1 Normal Operation

These are the results of the measurements with normal operation. Normal operation means that the master node and the slave node are connected to the same network and there is no heavy CPU load. These measurements were performed to fulfill the requirements R-RT07.

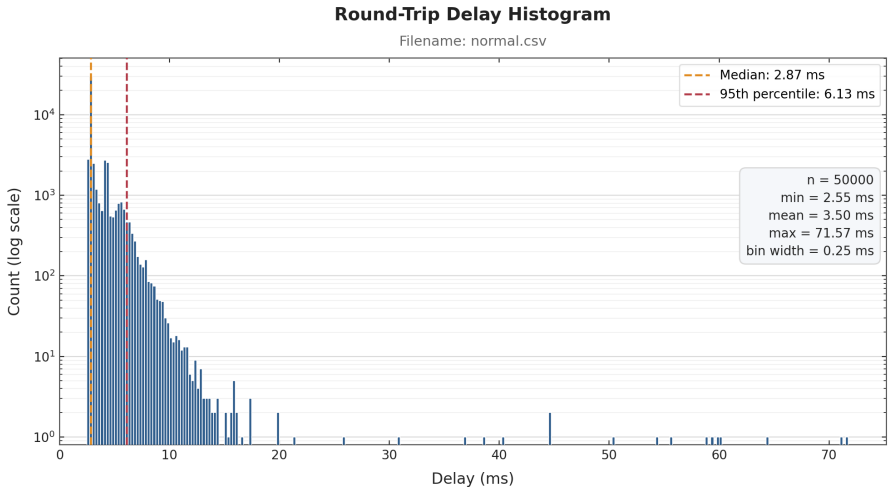


Abbildung 3: Measurement results for 50000 measurements under normal conditions.

2.2 High Network Load

These are the results of the measurements with high network load. To simulate a high network load the tool `iperf` was used in combination with another client which was a notebook connected to the same network. These measurements were performed to fulfill the requirements R-RT08.

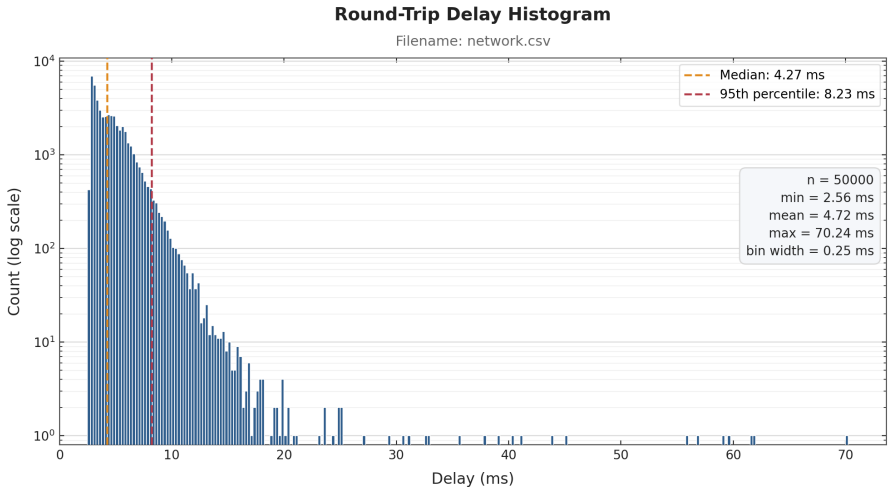


Abbildung 4: Measurement results for 50000 measurements under high network load.

2.3 High CPU Load

These are the results of the measurements with high network load. To simulate a high network load the tool `prime95` was used. This tool calculates prime numbers to generate load on the CPU. The tool was run on the master node. These measurements were performed to fulfill the requirements R-RT09.

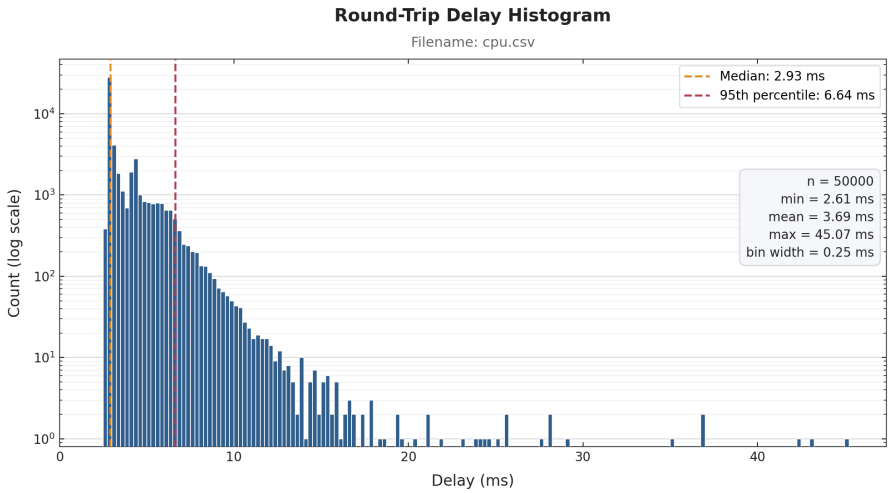


Abbildung 5: Measurement results for 50000 measurements under CPU load.

2.4 Overview Tables

2.4.1 Statistics

Tabelle 1: Round-trip delay statistics

Data set	Min [ms]	Mean [ms]	Median [ms]	p95 [ms]	Max [ms]
normal	2.550	3.497	2.870	6.132	71.566
cpu	2.608	3.695	2.928	6.640	45.066
network	2.563	4.721	4.271	8.227	70.245

2.4.2 Completeness

Tabelle 2: Round-trip measurement completeness

Data set	Sent	Received	Timeouts	Aborted	No valid response
normal	50000	50000	0	0	0/50000 (0.00 %)
cpu	50000	50000	0	0	0/50000 (0.00 %)
network	50708	50000	708	0	708/50708 (1.40 %)